

ProctorShield AI — System Test Document (STD)

System Test Plan, Automated Type Verification & System Test Case Execution Matrix

ProctorShield AI — Official System Documentation | Generated: August 2026

1. Test Strategy & Scope Overview

This document outlines the System Test Plan and Test Case Specifications for ProctorShield AI v1.0.0. The test suite evaluates core authentication, multi-tab video stream resilience, client-side AI computer vision, anti-cheat lockdown security, retake workflows, and database storage management.

2. Automated Build & Type Check Verification

Automated Compiler Verification Command:

```
$ npx tsc --noEmit
```

Execution Result: Passed clean with 0 TypeScript compilation errors. All API payload interfaces, Prisma queries, and component prop definitions are strictly typed.

3. Complete System Test Execution Matrix

Test Case ID	Module / Component	Test Scenario	Expected Outcome	Status
TC-AUTH-01	Authentication	Student & Teacher Login	JWT Cookie issued, redirected	PASS
TC-AUTH-02	Authentication	Multi-Tab Session Test	Payload studentId bypasses	PASS
TC-AUTH-03	Auth / Security	Forgot Password OTP	6-digit OTP dispatched & ver	PASS
TC-MON-01	Live Video	WebRTC 60 FPS Stream	P2P channel connects with 60	PASS
TC-MON-02	Live Video	1.5s Snapshot Fallback	Snapshot updates every 1.5s	PASS
TC-MON-03	Live Video	5s WebM Video Evidence	MediaRecorder captures 5s of	PASS
TC-AI-01	AI Proctoring	3D Head Direction Radar	Radar dot glides to UP/DOWN	PASS
TC-AI-02	AI Proctoring	Cell Phone Scanning	COCO-SSD detects phone & flag	PASS
TC-AI-03	AI Proctoring	3-Strike Auto Submit	Quiz terminates & submits on	PASS
TC-QUIZ-01	Quiz Lockdown	Fullscreen Enforcement	Enters fullscreen on quiz star	PASS
TC-QUIZ-02	Quiz Lockdown	5s Startup Grace Period	0 false violations during full...	PASS
TC-QUIZ-03	Quiz Lockdown	Pre-Warning Banner Delay	Amber warning toast appears	PASS
TC-ADM-01	Admin Portal	User Suspension	Account status updated to sus	PASS
TC-ADM-02	Admin Portal	CSV Log Export	Real CSV file generated & do	PASS
TC-ADM-03	Storage Mgmt	Neon DB Storage Purge	DELETE route purges eviden	PASS

4. Test Execution Sign-Off

System Test Sign-off:

- Total Test Cases Executed: 15
- Test Cases Passed: 15 (100% Pass Rate)
- Test Cases Failed: 0
- Defect Status: All critical, major, and minor defects resolved. System is verified for production deployment.